

Simultaneous equations and inequalities - Set 2 - Solutions

Specification MM1.4, MM1.5, M4.16, M4.17.

Answer key

Question	Answer	Topic
1	B	Simultaneous equations and inequalities
2	A	Simultaneous equations and inequalities
3	H	Simultaneous equations and inequalities
4	E	Simultaneous equations and inequalities
5	F	Simultaneous equations and inequalities
6	C	Simultaneous equations and inequalities
7	B	Simultaneous equations and inequalities

Worked solutions

Question 1 SIM-08 · Simultaneous equations and inequalities · spec MM1.5

Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?

Answer: B

If $x > 0$: $1 > 2x$, so $0 < x < \frac{1}{2}$. If $x < 0$: $\frac{1}{x}$ is negative and cannot exceed 2. So $0 < x < \frac{1}{2}$. (Multiplying by x without considering its sign is the classic error.)

Full method - how and why

An inequality with the variable in the denominator. You must not multiply by x without knowing its sign - split into cases instead.

- 1 Case $x > 0$: multiplying by x keeps the direction, giving $1 > 2x$, i.e. $x < \frac{1}{2}$. Combined with $x > 0$: $0 < x < \frac{1}{2}$.
why: Multiplying an inequality by a positive quantity is safe; the case assumption must be carried into the answer.
- 2 Case $x < 0$: then $\frac{1}{x} < 0 < 2$, so the inequality can never hold.
why: Rather than multiplying by a negative (and flipping), just observe the sign - a negative number is never greater than 2.
- 3 Case $x = 0$: $\frac{1}{x}$ is undefined, so $x = 0$ is excluded.
why: Always check the value that breaks the expression.
- 4 Combine: the solution set is exactly $0 < x < \frac{1}{2}$.
why: Only the first case contributes.

Where it goes wrong

- Multiplying through by x blindly to get $x < \frac{1}{2}$, which wrongly includes every negative number.
- Answering $x > 2$ by 'inverting' both sides without justification.

Takeaway Never multiply an inequality by an expression of unknown sign: split into cases, or bring everything to one side and analyse the sign of the fraction.

Question 2 SIM-10 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $(x - 1)(x - 3)(x - 5) > 0$.

Answer: A

Sign chart: the product is positive for $x > 5$, negative for $3 < x < 5$, positive for $1 < x < 3$, negative for $x < 1$.

Full method - how and why

A cubic inequality. With three distinct linear factors, use a sign chart - the sign alternates as you cross each root.

- 1 The roots are $x = 1, 3, 5$; mark them on a number line, splitting it into four regions.
why: A product of linear factors can only change sign at a root.
- 2 Test the far-right region, e.g. $x = 6$: $(5)(3)(1) > 0$.
why: One test value fixes the whole chart; the far right is easiest because every factor is positive.
- 3 Alternate leftwards: positive for $x > 5$, negative on $(3, 5)$, positive on $(1, 3)$, negative for $x < 1$.
why: Each simple (unrepeated) root flips the sign exactly once.
- 4 Collect the positive regions: $1 < x < 3$ or $x > 5$.
why: The answer is a union of two intervals.

Where it goes wrong

- Answering $x > 5$ only, forgetting the middle positive region.
- Assuming the expression is positive whenever x is bigger than all the roots - true here, but the alternation is what you must use.

Takeaway Sign chart: find the roots, test one region, then alternate; repeated factors are the only ones that do not flip the sign.

Question 3 SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that $|2x - 3| < 5$, which of the following statements is/are true?

I $-1 < x < 4$

II $x^2 < 16$

III $x > -1$

Answer: H

$-5 < 2x - 3 < 5 \Rightarrow -1 < x < 4$, so I is true and III follows from I. For II: every x in $(-1, 4)$ has $x^2 < 16$ (the largest possible $|x|$ is just under 4), so II is also true.

Full method - how and why

A modulus inequality plus three consequences. Solve the modulus first, then test each statement against that interval.

1 Unpack the modulus: $|2x - 3| < 5$ means $-5 < 2x - 3 < 5$.

why: $|A| < k$ means A lies within k of zero - a double inequality, not two separate cases.

2 Solve the double inequality: add 3 throughout to get $-2 < 2x < 8$, then halve: $-1 < x < 4$. So **I is true**.

why: Whatever you do to one part of a double inequality you do to all three parts.

3 III says $x > -1$, which is half of I. **True**.

why: A weaker statement implied by a true one is automatically true.

4 II: for $-1 < x < 4$, the largest $|x|$ can get is just under 4, so $x^2 < 16$ always. **True**.

why: For squares, what matters is the largest **absolute value** in the interval, not the largest value.

5 All three hold.

why: Match to the option list.

Where it goes wrong

- Treating $|2x - 3| < 5$ as $2x - 3 < 5$ only, losing the lower bound.
- Rejecting II by testing a value like $x = -3$ that is not in the interval at all.

Takeaway $|A| < k \Leftrightarrow -k < A < k$; when squaring an interval, check the end with the larger absolute value.

Question 4 SIM-13 · Simultaneous equations and inequalities · spec MM1.5

How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?

Answer: E

$x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$. $x(x - 3) > 0 \Rightarrow x < 0$ or $x > 3$. Integers: $-2, -1$ and $4, 5, 6, 7$: six.

Full method - how and why

Two quadratic inequalities, then an integer count. The second one factorises without a constant term, which changes its shape.

- 1 First: $x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$.
why: Between the roots for < 0 .
- 2 Second: $x^2 - 3x > 0 \Rightarrow x(x - 3) > 0 \Rightarrow x < 0$ or $x > 3$.
why: Roots 0 and 3; outside the roots for > 0 . Do not divide by x - that would lose the $x < 0$ branch.
- 3 Intersect: from $(-3, 8)$ keep $(-3, 0)$ and $(3, 8)$.
why: Overlap both solution sets on one number line.
- 4 Count integers: $-2, -1$ from the first piece and $4, 5, 6, 7$ from the second: six.
why: Endpoints $-3, 0, 3, 8$ are all excluded by strict inequalities.

Where it goes wrong

- Dividing $x^2 > 3x$ by x to get $x > 3$, losing all the negative solutions.
- Including 0 or 3 in the count.

Takeaway Never divide an inequality by the variable; factorise instead, then count integers explicitly.

Question 5 SIM-15 · Simultaneous equations and inequalities · spec MM1.5

The solution set of $ax^2 + bx + c < 0$ (with $a > 0$) is $-2 < x < 5$. What is the solution set of $ax^2 - bx + c < 0$?

Answer: F

Replacing x by $-x$ reflects the graph in the y -axis, so the roots -2 and 5 become 2 and -5 : $-5 < x < 2$.

Full method - how and why

A transformation question hidden in an inequality. Replacing x by $-x$ reflects the graph in the y -axis, and the solution set reflects with it.

- 1 Note that $a(-x)^2 + b(-x) + c = ax^2 - bx + c$; the second expression is the first with x replaced by $-x$.
why: Spotting this makes the question a one-liner; the alternative is to reconstruct a, b, c explicitly.
- 2 So the graph of the second is the reflection of the first in the y -axis.
why: $f(-x)$ is always the y -axis reflection of $f(x)$.
- 3 The original is negative on $-2 < x < 5$; reflecting sends x to $-x$, so the roots -2 and 5 move to 2 and -5 .
why: Every point of the solution set flips sign, and the interval flips with it.
- 4 The new solution set is $-5 < x < 2$.
why: Write the interval with the smaller number first.

Where it goes wrong

- Leaving the interval as $2 < x < -5$, which is empty as written.
- Assuming the solution set is unchanged because 'only the sign of b changed'.

Takeaway Changing b to $-b$ reflects a quadratic in the y -axis; everything about the graph, including solution sets, reflects too.

Question 6 SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.

Answer: C

A fraction is ≤ 0 when numerator and denominator have opposite signs (or the numerator is 0): $x - 2 \leq 0$ and $x + 1 > 0$ gives $-1 < x \leq 2$; $x - 2 \geq 0$ and $x + 1 < 0$ is impossible. $x = -1$ is excluded as the fraction is undefined.

Full method - how and why

A rational inequality. Never multiply by $x + 1$ (its sign is unknown) - analyse the sign of the fraction instead.

- Find the critical values: numerator zero at $x = 2$, denominator zero at $x = -1$.

why: The fraction can only change sign where the numerator or denominator vanishes.

- Test the three regions. For $x < -1$ (say $x = -2$): $\frac{-4}{-1} = 4 > 0$. For $-1 < x < 2$ (say $x = 0$): $\frac{-2}{1} = -2 < 0$. For $x > 2$ (say $x = 3$): $\frac{1}{4} > 0$.

why: One test value per region settles the sign throughout that region.

- Include $x = 2$ (the fraction is 0, and ≤ 0 allows it) but exclude $x = -1$ (undefined).

why: The two endpoints behave differently: a zero numerator is allowed by $\leq$; a zero denominator never is.

- Answer: $-1 < x \leq 2$.

why: Note the mixed brackets - strict at -1 , inclusive at 2.

Where it goes wrong

- Multiplying by $x + 1$ to get $x - 2 \leq 0$, i.e. $x \leq 2$, which wrongly includes everything below -1 .
- Writing $-1 \leq x \leq 2$ and including a value where the expression is undefined.

Takeaway For $\frac{A}{B} \leq 0$, use a sign chart on A and B ; a zero denominator is always excluded, a zero numerator is included when the inequality is not strict.

Question 7 SIM-18 · Simultaneous equations and inequalities · spec MM1.5

Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?

Answer: B

$x^2 - 3x + 2 = (x - 1)(x - 2) > 0$ exactly when $x < 1$ or $x > 2$. So $x > 2$ does imply it (true), but the converse fails, e.g. $x = 0$ gives $2 > 0$ without $x > 2$.

Full method - how and why

A Paper 2-style wrapper on a quadratic inequality: decide the statement and its converse separately.

- Factorise the quadratic: $x^2 - 3x + 2 = (x - 1)(x - 2)$, which is positive exactly when $x < 1$ or $x > 2$.

why: Knowing precisely where the expression is positive lets you test both directions quickly.

- The statement: does $x > 2$ guarantee positivity? Yes - $x > 2$ lies inside the region $x > 2$. **True.**

why: An implication is true when the hypothesis region sits inside the conclusion region.

- The converse: does positivity guarantee $x > 2$? No - $x = 0$ gives $(0 - 1)(0 - 2) = 2 > 0$ but $0 \not> 2$. **False.**

why: The positivity region is strictly larger than $x > 2$, so a counterexample must come from the extra part $x < 1$.

- So the statement is true but its converse is false.

why: Report both verdicts in the order the options use them.

Where it goes wrong

- Assuming the converse holds because the implication does - converses are logically independent.
- Only testing values greater than 2, which can never refute the converse.

Takeaway An implication is true when one region is contained in the other; its converse needs the reverse containment, so look for a counterexample in the leftover region.